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RHJ/UBM

19th February 1976.

PROJECT LIEUTENANT - CARBON MONOXIDE ANALYSIS.

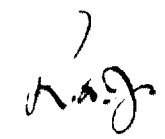
We refer to your memo of 16th February concerning the above. We have looked at the results for the UST brands covered by our report 76-02-027 and would comment as follows:-

1. The measurement we actually carry out on the smoke is the concentration of carbon monoxide (%) in the vapour phase. This figure is then used in a calculation to find the delivery. One of the terms used in this calculation is the puff number, since we need to know firstly the total volume of vapour phase delivered, which is puff number x puff volume (35c.c.) x number of cigarettes smoked. In fact the other terms in the calculation referred to are fairly constant and thus we can say that delivery at any particular concentration depends on puff number.

Thus, if we compare, say, Pall Mall Superlongs with Marlboro', both with a carbon monoxide concentration of 4.1% but deliveries of 18.5mg and 13.7mg. respectively, then the puff numbers of 11.1 and 8.2 respectively help us to explain the difference. In fact, on a simple delivery-per-puff basis, both cigarettes give 1.67mg.

2. It is probably more important to try to explain why Marlboro' gives a vapour phase with 4.1%CO, while Lucky Strike F/F gives 5.1%. One explanation would be the large difference in cigarette*porosity, Marlboro' at 501 and Lucky Strike at 285 and there may be others which are not covered by our examination. Nevertheless, as you say, these are early days, and many more samples would need to be tested before we would place the required degree of confidence in the results to say that there is a real difference and that we are not just seeing the effect of batch to batch variation.

* paper


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